Course Code: CSE3720	**Course Name:** Generative AI and LLMs	
Credits: 3 (3-0-0)	**Contact Hours: 3** Hours (approx.) (Theory + Lab) per week	
Batch: 2023, 6th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan,2026 to 8th May 2026	
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Aim of the Course:
The aim of this course is to provide learners with a structured and practical understanding of generative AI and large language models (LLMs). It begins with core foundational generative modeling concepts, including sequence-to-sequence models, variational autoencoders, and generative adversarial networks, and gradually progresses toward transformer-based architectures that underpin modern large language models. Building on this progression, the course introduces advanced topics such as prompt engineering, fine-tuning and parameter-efficient adaptation techniques, retrieval-augmented generation, and tool-enabled LLM workflows.
The course aims to equip students with a strong theoretical foundation and practical skills to design, adapt, deploy, fine-tune, and evaluate generative AI and LLM-based solutions for real-world applications. Ethical considerations, data privacy, and responsible AI practices are integral components of the course, ensuring that students develop safe, accountable, and effective generative AI systems suitable for real-world deployment.

Course Overview and Context:
This course offers a comprehensive journey through generative AI and LLMs, combining theoretical knowledge with extensive lab-based practice. The course covers key generative models, including sequence-to-sequence architectures, variational autoencoders, and generative adversarial networks, before progressing to transformer-based models and large language models. Students learn practical techniques for working with LLMs, such as prompt engineering, parameter-efficient fine-tuning, retrieval-augmented generation, and tool-enabled workflows using frameworks like LangChain. The curriculum also addresses data privacy, local deployment considerations, model evaluation challenges, and ethical issues related to bias, misinformation, and responsible use. With a strong emphasis on applied learning, the course prepares students for industry roles and research pathways in generative AI and LLM-based systems.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Apply core Large Language Model (LLM) concepts and effective prompting	Employability	→ Interactive theory sessions on LLM basics and prompting

	techniques to solve real-world problems		→ In-class tool demonstrations (ChatGPT, Copilot, Gemini) → Industry use-case studies and discussion
CO2	Analyze the Generative AI lifecycle and model architectures to determine their suitability for different application domains.	Skill Development	→ Comparative evaluation of models such as VAEs, GANs & Transformer → Implementation walkthroughs using Python libraries (TensorFlow/Keras/ PyTorch) → Mini assignments: Choose the best architecture for a use case & justify why.
CO3	Evaluate ethical and responsible AI frameworks and apply them in the design and implementation of safe, accountable, and effective Generative AI solutions.	Entrepreneurship + Skill Development	→ Case-studies on AI ethics, hallucinations, fairness issue → Building small working prototypes using LangChain, RAG, LoRA, etc. → Getting familiar with cloud services (HuggingFace Hub, Blynk/Gradio demo, OpenAI endpoints, etc.) → Capstone/mini-project: Build & demonstrate an LLM-powered solution relevant to industry or startup use

Prerequisite: if any

1. Python programming skills
2. Fundamentals of Machine Learning

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Unit 1: Generative AI Overview: What is generative AI, Different types of generative AI models, Understand the principles and applications of Generative AI in creating new data instances, Understand the principles behind Generative AI: Get familiar with building and tweaking generative models for any real-world use case. Generative AI Project Lifecycle.	2 sessions 2 Theory	CO1
Unit 2: Generative AI Models: Seq2Seq Models and Variational Autoencoders (VAEs): Sequence-to-Sequence models, Variational Autoencoders, Text-to-Text generation, Image and text-based VAE applications. Generative Adversarial Networks (GANs): Understanding GAN architecture, GAN training process, GAN applications in image and video generation.	6 sessions 2 Theory + 4 Lab	CO1
Unit 3: Transformer: Learn about the architecture of Transformer models, including attention mechanisms, encoders, and decoders, Pre-training and fine-tuning strategies. Gain deeper insights into the capabilities and potential of Transformer technology.	6 sessions 2 Theory + 4 Lab	CO1, CO2

Unit 4: Large Language Models: Learn about Language Models (LLMs) and their role in understanding and generating human-like text, Different types of LLMs: Small Language Models (SLMs), Dense Language Models, etc. Different training datasets for LLMs. Training methods for LLMs, Challenges in training LLMs, Different evaluation metrics for LLMs, Challenges in evaluating LLMs, Practical Implementation of LLMs, and how fine-tuning enables LLMs to be adapted to a variety of specific use cases. Data privacy and local LLMs. Introduction to LangChain for building LLM applications.	10 sessions 4 Theory + 6 Lab	CO1, CO2
Unit 5: Prompts Module: Understanding the concept and significance of prompt engineering, strategies for designing effective prompts, Techniques for controlling model behaviour and output quality. Prompting Techniques: Reasoning and Acting (ReAct), Chain-of-Thought (CoT).	6 sessions 2 Theory + 4 Lab	CO1, CO2
Unit 6: LLM Fine-Tuning Techniques Fine-tuning and Parameter-Efficient Fine-Tuning (PEFT), Low Rank Adaptation (LoRA), Reinforcement Learning with Human Feedback (RLHF).	6 sessions 2 Theory + 4 Lab	CO1, CO2, CO3
Unit 7: Retrieval-Augmented & Tool-Enhanced LLM Workflows Retrieval-Augmented Generation (RAG) – vector database and similarity search, Implementing RAG with LangChain. Overview of recently popular models (GPT-4/5, Llama-3, Mistral, and Gemini etc.), Use of Tools and function calling LLMs.	6 sessions 2 Theory + 4 Lab	CO1, CO2, CO3
Unit 8: Ethical Considerations & Responsible AI Practices: Potential biases in generative AI models, Misinformation and disinformation generated by LLMs and Other ethical concerns. Addressing bias and fairness in generative AI systems. Guardrails and ensuring responsible use of generative AI and deployment of generative models.	6 sessions 2 Theory + 4 Lab activity/case study	CO1, CO2, CO3

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	2	2					3		3	3	3	3	
CO2	2	3	2	3	2					2		3	3	3	2	
CO3	2	2	2	2	2	3	3	3		2	3	3	3	3	2	

 # Learning Resources:

Textbooks:

1. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play" by David Foster, O'Reilly Media. Second edition: June 6, 2023 (ISBN: 978-1098134181).
2. Generative AI with LangChain: Build large language model (LLM) apps with Python, ChatGPT, and other LLMs" by Ben Auffarth, Packt Publishing, first edition: December 22, 2023 (ISBN-13: 978-1835083468)
3. Build a Large Language Model (From Scratch)" by Sebastian Raschka, Manning Publications, 1st Edition, ISBN-13: 978-1633437166
4. Language Models: From Scratch to GPT-3" by Gabriel Garcia, Packt Publishing. ISBN: 978-1801815861
5. Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT, Press, 2016.
6. Deep Learning with Python by François Chollet, 2nd Ed., Manning.

Reference Books

1. Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention is all you need." Advances in neural information processing systems 30 (2017). [Research Paper]
2. Programming PyTorch for Deep Learning – Ian Pointer, O'Reilly, 2020 Covers PyTorch basics, which is essential for implementing Transformers and generative *AI models.*
3. Yenduri, Gokul, Gautam Srivastava, Praveen Kumar Reddy Maddikunta, Rutvij H. Jhaveri, Weizheng Wang, Athanasios V. Vasilakos, and Thippa Reddy Gadekallu. "Generative Pre-trained Transformer: A Comprehensive Review on Enabling Technologies, Potential Applications,

Video lectures/ MOOC:

Reference Links:
1. https://www.coursera.org/learn/generative-ai-llm-architecture-data-preparation

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule/Timeline	Rubrics/Remarks	CO mapping
Quiz	20%	April Last Week	-MCQ knowledge and programming-based questions. -Topics to be covered will be announced in the class. -20 MCQ questions, each of 1 mark. -Quiz duration 20-30 mins each	CO1, CO2
Assignment 1 & 2	2 x 20% = 40%	Continuous Assessment	Programming Assignments will be given. Offline evaluation to be conducted. Students will be asked to explain the solution with a **demo** followed by **viva**.	CO1, CO2

End-Sem Project Evaluation	40%	Last week of the course	A comprehensive **Project report** is to be submitted along with the coding solutions; this will be followed by **Presentations, Demonstrations & Viva**.	CO1, CO2, CO3
– All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. – Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity.				–

Experiential Learning Component (80%):

The course includes two programming assignments, each carrying 20% weightage (total 40%), aimed at developing practical skills in implementing and applying Generative AI and LLM concepts. In addition, the project component also serves as one of the experiential learning activities for this course. Students will work on real-world problem statements requiring the design and implementation of Generative AI and LLM-based solutions, such as prompt engineering, retrieval-augmented generation, model adaptation, or evaluation of generative outputs. Students are expected to implement the solution, analyze results, and assess model behaviour with respect to quality and ethical considerations. The project carries a 40% weightage, with an expected effort of 20+ hours, and will be evaluated separately. Collectively, these components emphasize hands-on, application-driven learning, with 80% of the overall course assessment dedicated to experiential learning.

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: This is a practical course with a lot of hands-on activities. There will be no recourse examination. Students failing in the course must repeat the course when it is offered next.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expect: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.